

DARSHIL VATYANI

Indore

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EDUCATION

B.Tech. in Information Technology

Shri G.S. Institute of Technology & Science, Indore

2023 – 2027

CGPA: 8.53/10

Coursework: Data Structures & Algorithms, Object Oriented Programming, Software Engineering, Operating Systems, Database Management Systems, Computer Networks

EXPERIENCE

Monarch & Mirth

Full Stack Intern

May 2026 – July 2026

- Developed and maintained 7+ REST endpoints using Node.js/Express.js with MongoDB for data persistence.
- Built responsive UI components in React.js, collaborating with the team on feature development and bug fixes.

PROJECTS

déjà – Record & Replay Debugger for Node.js

GitHub

- Engineered a record-and-replay debugger that captures every nondeterministic input of a live Node.js service (network I/O, timers, randomness, clocks, crypto, DNS, signals) and reproduces a rare production crash **offline, at the exact same event, on every replay**.
- Designed a deterministic scheduler on a virtual clock, replaying a 479-event incident in **0.13s (5× faster than real time)**, validated by fuzzing random async programs.
- Intercepted networking with fake sockets covering HTTP, fetch and raw TCP with **zero dependencies**; traced and eliminated nondeterministic connection ordering in Node's HTTP server.
- Added time-travel debugging via the **V8 Inspector Protocol** to inspect the stack and locals at any event, plus divergence detection pinpointing the first deviating line of code.
- **Tools Used:** Node.js, JavaScript, V8 Inspector Protocol, async_hooks, zstd, Fastify, HTML Canvas

qopt - PostgreSQL Query Advisor

- Built a PostgreSQL query advisor that captures slow queries and diagnoses bottlenecks from execution plans.
- Combined rule-based checks with Gemini and hybrid documentation search to suggest indexes and SQL rewrites.
- Validated fixes with HypoPG and a shadow database, checking query results and execution time; rolled back experiments and sent rejection reasons to Gemini for revision.
- Compared four configurations on a 15-query demo workload; validation with documentation retrieval cut total execution time from **1,086 ms to 190 ms (5.7x faster)**.
- **Tools Used:** TypeScript, Node.js, Fastify, React, PostgreSQL, Gemini API, pgvector, HypoPG

SKILLS

Languages

C, C++, Python, JavaScript, SQL

Tools & Frameworks

Git, GitHub, Docker, Postman, NumPy, Pandas, Scikit-learn, Langchain, Socket.IO

Databases

MySQL, SQLite, MongoDB

Web Development

HTML, CSS, React.js, Node.js, Express.js, REST APIs, LLM APIs

ACHIEVEMENTS & CERTIFICATIONS

Solved **500+ DSA problems** on LeetCode & GeeksforGeeks and achieved max rating of 1675 on LeetCode.

Python for Data Science (NPTEL) – NumPy, Pandas, data visualisation, and basic ML using scikit-learn.

OCI 2025 Certified AI Foundations Associate – Foundational AI and ML knowledge on Oracle Cloud.

Finalist at **SRIJAN 2025** (RGPV Innovation Platform), one of 150 projects selected across MP.